

CHAPTER 14

Electromagnetic Waves in Free Space

1. THE WAVE EQUATION

Plasmas are able to sustain a great variety of *wave phenomena*. Before we get started in the study of wave phenomena in plasmas, we review in this chapter some of the basic features of electromagnetic waves propagating in free space. The starting point for deriving the partial differential equation for electromagnetic waves in free space is Maxwell equations, which for free space ($\rho = 0$ and $\vec{J} = 0$) may be written as

$$\vec{\nabla} \cdot \vec{E} = 0 \quad (1.1)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (1.2)$$

$$\vec{\nabla} \times \vec{E} = -\partial \vec{B} / \partial t \quad (1.3)$$

$$\vec{\nabla} \times \vec{B} = (1/c^2) \partial \vec{E} / \partial t \quad (1.4)$$

Taking the time derivative of both sides of (1.4), yields

$$\vec{\nabla} \times (\partial \vec{B} / \partial t) = (1/c^2) \partial^2 \vec{E} / \partial t^2 \quad (1.5)$$

Substituting $\partial \vec{B} / \partial t$ from (1.3), gives

$$-\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = (1/c^2) \partial^2 \vec{E} / \partial t^2 \quad (1.6)$$

Using the vector operator identity

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} \quad (1.7)$$

and noting that $\vec{\nabla} \cdot \vec{E} = 0$ in free space, we obtain

$$\nabla^2 \vec{E} - (1/c^2) \partial^2 \vec{E} / \partial t^2 = 0 \quad (1.8)$$

In a similar way we can perform the same operations on (1.3) to obtain

$$\nabla^2 \vec{B} - (1/c^2) \partial^2 \vec{B} / \partial t^2 = 0 \quad (1.9)$$

Eqs. (1.8) and (1.9) are the *vector* wave equations satisfied by the electromagnetic field vectors $\vec{E}$ and $\vec{B}$ in free space. The velocity of propagation of such waves is $c = 1/\sqrt{\mu_0 \epsilon_0}$. Since these equations are satisfied by each component of the field vectors $\vec{E}$ and $\vec{B}$, we may as well write a *scalar* wave equation

$$\nabla^2 \psi - (1/c^2) \partial^2 \psi / \partial t^2 = 0 \quad (1.10)$$

where ψ is used to denote any one of the components of $\vec{E}$ and $\vec{B}$.

2. SOLUTION IN PLANE WAVES

We are interested in *transverse plane wave* solutions of the partial differential wave equation (1.10), since these are the simplest and most fundamental electromagnetic waves. For transverse plane waves, the field vectors $\vec{E}$ and $\vec{B}$ lie on a plane perpendicular to the propagation direction and are functions only of the perpendicular distance from the origin to this plane and, of course, of time also. This plane, normal to the propagation direction, is called the *wave front*. Let z denote the perpendicular distance from the origin to the wave front plane and let $\vec{k}$ be a vector normal to this plane (Fig. 1). Any point P, in the wave front, can be represented by a position vector $\vec{r}$ drawn from the origin of the coordinate system. Thus, for any point in a given wave front we have $\vec{k} \cdot \vec{r} = z = \text{constant}$, which is the equation specifying this wave front plane. The direction cosines of the unit vector $\vec{k}$, in a Cartesian coordinate system, are given by $\vec{k} \cdot \vec{x}$, $\vec{k} \cdot \vec{y}$ and $\vec{k} \cdot \vec{z}$.

Since the field vectors $\vec{E}$ and $\vec{B}$ are spatially constant along the wave front (normal to $\vec{k}$), but vary only in the z direction (and with time), the del operator can be written as

$$\vec{\nabla} = \hat{x} \partial/\partial x + \hat{y} \partial/\partial y + \hat{z} \partial/\partial z = \vec{k} \partial/\partial z \quad (2.1)$$

Hence, the wave equation is *one-dimensional* in form for the variable z ,

$$\partial^2 \psi / \partial \zeta^2 - (1/c^2) \partial^2 \psi / \partial t^2 = 0 \quad (2.2)$$

The general solution of this one-dimensional wave equation is a combination of arbitrary functions of the variables $(\zeta - ct)$ and $(\zeta + ct)$,

$$\psi(\zeta, t) = f(\zeta - ct) + g(\zeta + ct) \quad (2.3)$$

The function $f(\zeta - ct)$ represents a plane waveform propagating in the positive ζ direction, whereas $g(\zeta + ct)$ corresponds to a waveform propagating in the negative ζ direction, the velocity of propagation being $c = 1/\sqrt{\mu_0 \epsilon_0}$.

3. HARMONIC WAVES

A particularly important type of plane waveform is the harmonic wave, which has the form (for propagation in the positive ζ direction)

$$\psi(\zeta, t) = A \cos [k(\zeta - ct)] = A \cos (k\zeta - \omega t) \quad (3.1)$$

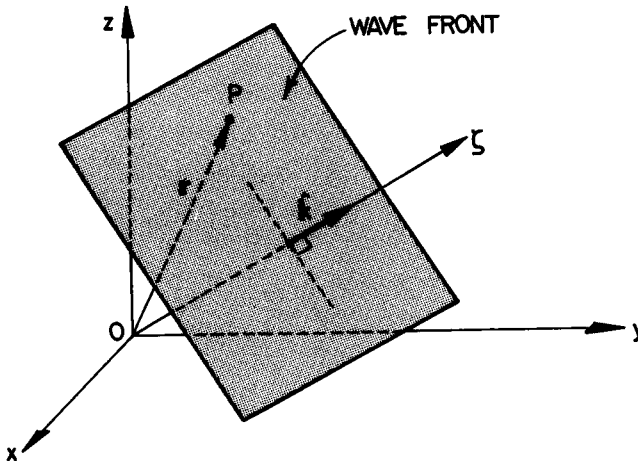


Fig. 1 - In a transverse plane wave the field vectors do not vary along a given wave front plane, normal to the direction of propagation ζ . They vary only along the ζ direction and with time.

where $\omega = kc$ is the *angular frequency* of the oscillation and k is the *wave number* or *propagation constant*. Fig. 2 shows the space and time dependence of the harmonic wave $\psi(\zeta, t)$ given by (3.1). The wavelength λ and the period T of the wave motion are given, respectively, by

$$\lambda = 2\pi/k ; \quad T = 2\pi/\omega = 1/\nu \quad (3.2)$$

where ν is the frequency in cycles per second.

If we define a *propagation vector* $\vec{k}$, whose direction is that of the normal to the wave front ($\vec{k}$) and whose magnitude is the wave number ($|\vec{k}| = k$), then, since ζ is the perpendicular distance from the origin to the wave front, we have

$$\vec{k} \cdot \vec{r} = (\vec{k}/k) \cdot \vec{r} = \zeta \quad (3.3)$$

Hence, for a harmonic plane wave travelling in some arbitrary direction specified by the unit vector $\vec{k}$,

$$\psi(\vec{r}, t) = A \cos (\vec{k} \cdot \vec{r} - \omega t) \quad (3.4)$$

In view of the argument of the cosine function (3.4), the planes of *constant phase* are defined by the condition

$$\vec{k} \cdot \vec{r} - \omega t = k\zeta - \omega t = \text{constant} \quad (3.5)$$

The *phase velocity* is defined as the velocity of propagation of these planes of constant phase ($d\zeta/dt$) and is found by differentiating (3.5) with respect

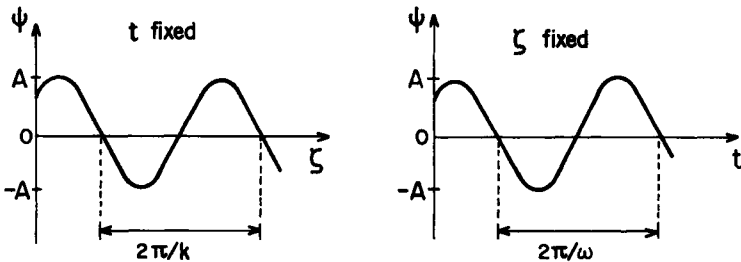


Fig. 2 - Amplitude of a harmonic plane wave as a function of space and time.

to time,

$$v_{ph} = \omega/k \quad (3.6)$$

The phase velocity is positive for a wave moving in the positive z direction, that is, z increases as t increases in order to keep $kz - \omega t$ constant. If we had taken

$$\psi(\vec{r}, t) = A \cos(\vec{k} \cdot \vec{r} + \omega t) \quad (3.7)$$

then the phase velocity would be negative for a wave moving in the negative z direction.

It is generally convenient and extremely useful to write (3.4) in complex form

$$\psi(\vec{r}, t) = A \exp[i(\vec{k} \cdot \vec{r} - \omega t)] \quad (3.8)$$

where implicit in this expression is the understanding that the field quantities are obtained by taking the real part of (3.8). The true physical quantity involved (the one we would measure in an experiment) is the real part of the complex form. The use of complex expressions, however, simplifies the mathematical calculations and, at the end, we take the real part of the result. This can be done as long as the equations are linear.

Using the complex notation (3.8) for the harmonic plane wave solutions for the field vectors $\vec{E}$ and $\vec{B}$, the operators $\vec{\nabla}$ and $\partial/\partial t$ become

$$\vec{\nabla} = i \vec{k} \quad ; \quad \partial/\partial t = -i\omega \quad (3.9)$$

so that Maxwell equations (1.1) to (1.4) in free space become simple (linear, homogeneous) algebraic equations

$$\vec{k} \cdot \vec{E} = 0 \quad (3.10)$$

$$\vec{k} \cdot \vec{B} = 0 \quad (3.11)$$

$$\vec{k} \times \vec{E} = \omega \vec{B} \quad (3.12)$$

$$\vec{k} \times \vec{B} = -(\omega/c^2) \vec{E} \quad (3.13)$$

Thus, from (3.10) and (3.11) we see that $\vec{E}$ and $\vec{B}$ are both perpendicular to $\vec{k}$ and, for this reason, these waves are called *transverse waves*. In addition, from (3.12) or (3.13) we see that $\vec{E}$ and $\vec{B}$ are also perpendicular to each other. This situation is illustrated in Fig. 3. The set of vectors $(\vec{E}, \vec{B}, \vec{k})$, taken in this order, constitute a right-handed orthogonal set. Fig. 4 illustrates the relation between the $\vec{E}$ and $\vec{B}$ vectors of a plane electromagnetic wave propagating in free space.

4. POLARIZATION

From (3.8) it is seen that the field vectors $\vec{E}$ and $\vec{B}$, which are solutions of the wave equation in the form of harmonic plane waves propagating in the positive z direction, can be written as

$$\vec{E}(\vec{r}, t) = \vec{E}_0 \exp [i(\vec{k} \cdot \vec{r} - \omega t)] = \vec{E}_0 \exp [i(kz - \omega t)] \quad (4.1)$$

$$\vec{B}(\vec{r}, t) = \vec{B}_0 \exp [i(\vec{k} \cdot \vec{r} - \omega t)] = \vec{B}_0 \exp [i(kz - \omega t)] \quad (4.2)$$

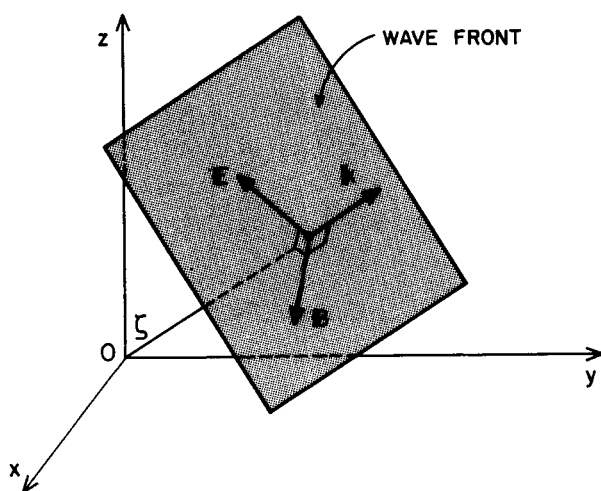


Fig. 3 - The propagation vector $\vec{k}$ and the two field vectors $\vec{E}$ and $\vec{B}$ are orthogonal to one another.

where $\vec{E}_0$ and $\vec{B}_0$ are constant vector amplitudes, which may be complex quantities. The phenomenon of polarization can be discussed entirely from the point of view of the electric field vector, since the magnetic induction vector may always be obtained from $\vec{E}$ using (3.12) for the case of plane waves.

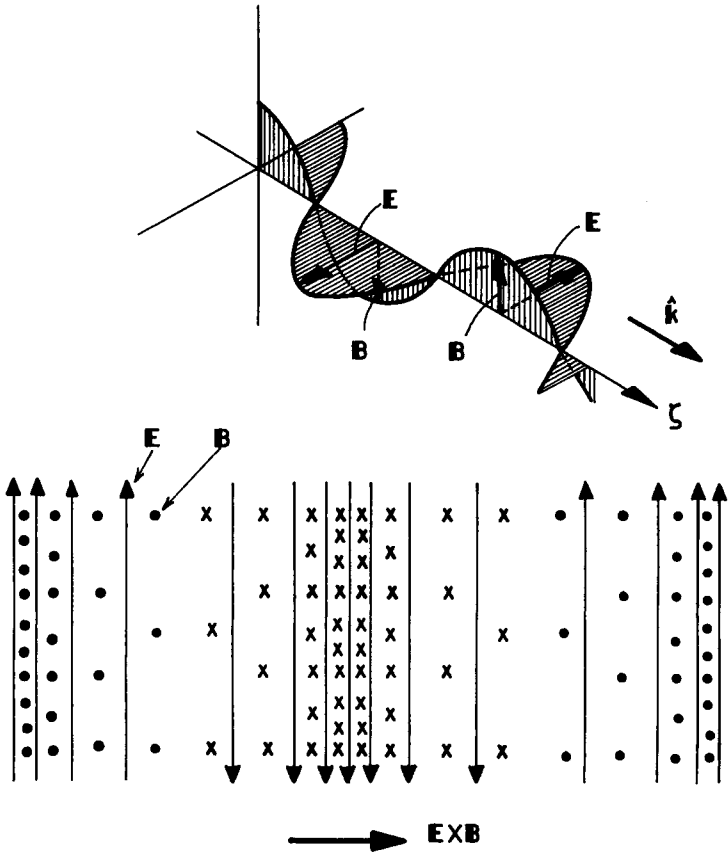


Fig. 4 - The field vectors $\vec{E}$ and $\vec{B}$ of a plane electromagnetic wave in free space. The dots and crosses represent magnetic lines of force coming out of the paper and into the paper, respectively, and the vertical lines represent the electric field. The direction of propagation is given by $\hat{k}$ which, in free space, is along $\vec{E} \times \vec{B}$.

According to (4.1), the direction of the electric field vector $\vec{E}$ is always the same, and the wave is said to be *linearly polarized*. This is the simplest type of polarization state.

In general, in order to describe an arbitrary state of polarization, we must consider the electric field vector in a plane as the superposition of two linearly independent, linearly polarized waves. Two such linearly independent waves can be represented by

$$\vec{E}_1(\vec{r}, t) = \hat{e}_1 E_1 \exp [i(\vec{k} \cdot \vec{r} - \omega t)] \quad (4.3)$$

$$\vec{E}_2(\vec{r}, t) = \hat{e}_2 E_2 \exp [i(\vec{k} \cdot \vec{r} - \omega t)] \quad (4.4)$$

with the associated magnetic induction vectors given, respectively, by

$$\vec{B}_1(\vec{r}, t) = (1/\omega) \vec{k} \times \vec{E}_1(\vec{r}, t) \quad (4.5)$$

$$\vec{B}_2(\vec{r}, t) = (1/\omega) \vec{k} \times \vec{E}_2(\vec{r}, t) \quad (4.6)$$

The unit vectors $\hat{e}_1$ and $\hat{e}_2$, called the *polarization vectors*, are perpendicular to each other and lie in the plane normal to the propagation vector $\vec{k}$, and are such that $(\hat{e}_1, \hat{e}_2, \vec{k})$, taken in this order, form a right-handed orthogonal set of unit vectors (Fig. 5). The amplitudes E_1 and E_2 are complex numbers, in

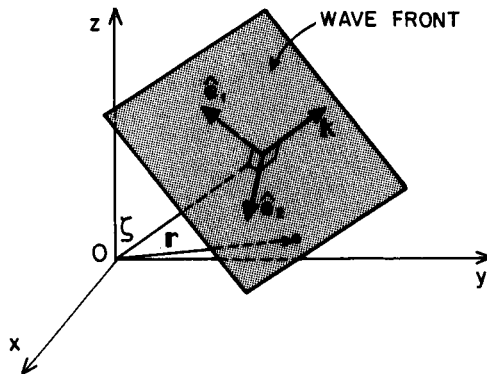


Fig. 5 - The two orthogonal polarization vectors $\hat{e}_1$ and $\hat{e}_2$, and the propagation vector $\vec{k}$.

order to allow for the possibility of phase difference between the two waves $\vec{E}_1(\vec{r}, t)$ and $\vec{E}_2(\vec{r}, t)$. Therefore, a general solution for a harmonic plane wave propagating in the direction $\vec{k}$ can be written as a linear combination of the two linearly independent solutions $\vec{E}_1$ and $\vec{E}_2$,

$$\vec{E}(\vec{r}, t) = (\hat{e}_1 E_1 + \hat{e}_2 E_2) \exp [i(\vec{k} \cdot \vec{r} - \omega t)] \quad (4.7)$$

Since any complex quantity can be expressed as the product of a real quantity and a complex phase factor, we may write

$$E_1 = E_1^0 \exp (i\alpha_1) \quad (4.8)$$

$$E_2 = E_2^0 \exp (i\alpha_2) \quad (4.9)$$

where the amplitudes E_1^0 and E_2^0 are real numbers, and α_1 and α_2 represent the phase of the complex amplitudes E_1 and E_2 , respectively. Note that it is only the *phase difference* that is physically significant and not the absolute magnitudes of α_1 and α_2 . Hence, (4.7) becomes

$$\vec{E}(\vec{r}, t) = [\hat{e}_1 E_1^0 \exp (i\alpha_1) + \hat{e}_2 E_2^0 \exp (i\alpha_2)] \exp [i(\vec{k} \cdot \vec{r} - \omega t)] \quad (4.10)$$

If E_1 and E_2 have the *same phase* ($\alpha_1 = \alpha_2$), then (4.10) represents a *linearly polarized wave*. The magnitude of $\vec{E}$ is

$$E = (E_1^2 + E_2^2)^{1/2} \quad (4.11)$$

and its polarization vector makes an angle

$$\theta = \tan^{-1} (E_2/E_1) \quad (4.12)$$

with the direction of $\hat{e}_1$. The electric field of this linearly polarized wave is represented in Fig. 6.

If E_1 and E_2 have *different phases* ($\alpha_1 \neq \alpha_2$), then (4.10) represents, in general, an *elliptically polarized wave*. The most simple case is the one in which the amplitudes of the components are equal, $E_1^0 = E_2^0 = E^0$, but the phases differ by $\pi/2$, that is, $\alpha_1 = 0$ and $\alpha_2 = \pm \pi/2$. Then, the wave (4.10) becomes [since $\exp (\pm i\pi/2) = \pm i$],

$$\vec{E}(\vec{r}, t) = E^0 (\hat{e}_1 \pm i\hat{e}_2) \exp [i(\vec{k} \cdot \vec{r} - \omega t)] \quad (4.13)$$

This wave is said to be *circularly polarized*. In order to illustrate this point, consider a Cartesian coordinate system such that the wave is propagating in the z direction, $\vec{k} = \hat{z}$, with $\hat{e}_1 = \hat{x}$ and $\hat{e}_2 = \hat{y}$. Then, taking the *real part* of (4.13) we obtain for the x and y components of the *actual* electric field,

$$\vec{E}_x(z, t) = E^0 \hat{x} \cos (kz - \omega t) \quad (4.14)$$

$$\vec{E}_y(z, t) = \mp E^0 \hat{y} \sin (kz - \omega t) \quad (4.15)$$

At a fixed plane $z = \text{constant}$, the fields are such that the $\vec{E}$ vector has a constant magnitude E^0 , but rotates around in a circle at the frequency ω . For the upper sign ($\hat{e}_1 + i\hat{e}_2$) the direction of rotation is *clockwise* if the wave is observed by looking at the outgoing wave front (i.e. looking along the positive z direction), as shown in Fig. 7 (a). Such a wave is said to be *right circularly polarized*. For the lower sign ($\hat{e}_1 - i\hat{e}_2$) the rotation is in the *counterclockwise* direction as shown in Fig. 7 (b), and the wave is said to be *left circularly polarized*.

In the most general case the amplitudes are different ($E_1^0 \neq E_2^0$) and the fields $\vec{E}_1$ and $\vec{E}_2$ have an arbitrary phase difference ($\alpha_1 \neq \alpha_2$). Considering

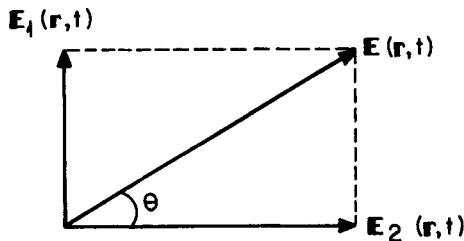


Fig. 6 - Electric field vector of a linearly polarized wave, represented by the superposition of two independent, linearly polarized waves $\vec{E}_1$ and $\vec{E}_2$, having the same phase.

$(\hat{e}_1, \hat{e}_2, \hat{k}) = (\hat{x}, \hat{y}, \hat{z})$, we have, from (4.10),

$$\vec{E}_x(z, t) = \hat{x} E_x^0 \exp(i\alpha_1) \exp[i(kz - \omega t)] \quad (4.16)$$

$$\vec{E}_y(z, t) = \hat{y} E_y^0 \exp(i\alpha_2) \exp[i(kz - \omega t)] \quad (4.17)$$

Taking the *real part* of these expressions we obtain

$$E_x(z, t) = E_x^0 \cos(kz - \omega t + \alpha_1) \quad (4.18)$$

$$E_y(z, t) = E_y^0 \cos(kz - \omega t + \alpha_2) \quad (4.19)$$

Squaring and adding (4.18) and (4.19), yields

$$\left(\frac{E_x}{E_x^0} \right)^2 + \left(\frac{E_y}{E_y^0} \right)^2 = \cos^2(kz - \omega t + \alpha_1) + \cos^2(kz - \omega t + \alpha_2) \quad (4.20)$$

This result shows that at a fixed plane $z = \text{constant}$ the electric field vector

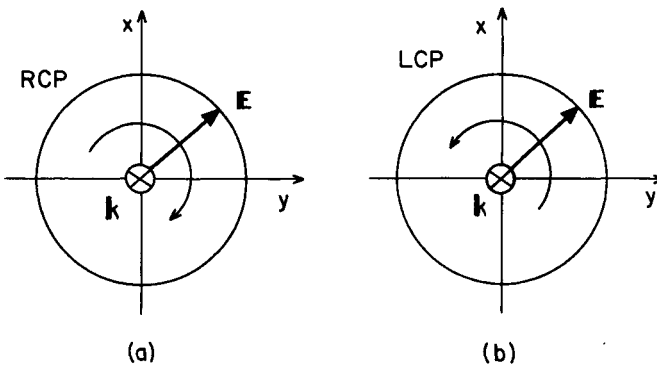


Fig. 7 - Electric field vector of a right circularly polarized wave (a), and a left circularly polarized wave (b). The propagation vector $\vec{k}$ points into the paper.

$\vec{E}$ performs, in general, an elliptical motion as a function of time.

For the case when $\alpha_1 = 0$ and $\alpha_2 = \pi/2$, (4.20) becomes, at the plane $z = 0$,

$$\left(\frac{E_x}{E_x^0} \right)^2 + \left(\frac{E_y}{E_y^0} \right)^2 = \cos^2(-\omega t) + \cos^2(-\omega t + \pi/2) = 1$$

since for any angle ϕ we have $\cos(\phi + \pi/2) = -\sin(\phi)$. In this case, the x and y components of the electric field are given by

$$E_x(0,t) = E_x^0 \cos(\omega t) \quad (4.22)$$

$$E_y(0,t) = E_y^0 \sin(\omega t) \quad (4.23)$$

Therefore, the tip of the electric field vector traces an ellipse in the $x - y$ plane rotating in the *clockwise* direction (*right-hand polarization*) when the wave is propagating away from the observer, as illustrated in Fig. 8.

For the case when $\alpha_1 = 0$ and $\alpha_2 = -\pi/2$, the electric field vector rotates in the *counterclockwise* direction when the wave is propagating away from the

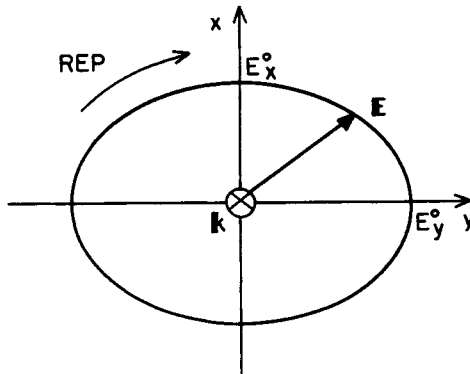


Fig. 8 - Electric field vector of a right-hand elliptically polarized wave ($\alpha_1 = 0$, $\alpha_2 = \pi/2$), rotating in the clockwise direction for an observer looking at the outgoing wave ($\vec{k}$ points into the paper).

observer, and the wave is said to be *left-hand elliptically polarized*, as shown in Fig. 9.

The ellipticity of the ellipse traced and the orientation of the major axis with respect to the x axis, depend on the relative magnitude of the amplitudes E_x^0 and E_y^0 , and on the phase difference $(\alpha_2 - \alpha_1)$. Fig. 10 illustrates various wave polarizations, obtained from (4.18) and (4.19) at the plane $z = 0$, for the case when $E_x^0 = E_y^0$. The angle ϕ indicates the phase difference, $\phi = (\alpha_2 - \alpha_1)$. In the figures shown, the sense of rotation of the $\vec{E}$ vector in the $x - y$ plane depends on whether ϕ lies between 0 and π , or between π and 2π .

5. ENERGY FLOW

Associated with the electric and magnetic field vectors $\vec{E}$ and $\vec{H}$ of an electromagnetic wave, there is a *flow of energy* (energy per unit area and per unit time) in the direction perpendicular to both $\vec{E}$ and $\vec{H}$. This energy flow is given the *Poynting vector*

$$\vec{S} = \vec{E} \times \vec{H} \quad (5.1)$$

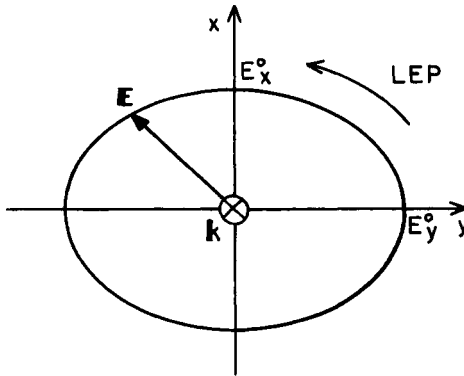


Fig. 9 - Electric field vector of a left-hand elliptically polarized wave ($\alpha_1 = 0, \alpha_2 = -\pi/2$), rotating in the counterclockwise direction for an observer looking at the outgoing wave ($\vec{k}$ points into the paper).

In free space $\vec{H}$ and $\vec{B}$ are related by $\vec{B} = \mu_0 \vec{H}$. Also, in free space the field vectors $\vec{E}$ and $\vec{H}$, and the propagation vector $\vec{k}$, form a mutually orthogonal set of vectors, but this is not the case in a conducting media as, for example, in a plasma. In such a medium, because of the presence of polarization charges, Maxwell equation $\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$ requires that $\vec{k} \cdot \vec{E} \neq 0$. Nevertheless, the direction of energy flow, given by the Poynting vector, is always perpendicular to $\vec{E}$ and $\vec{H}$.

When the field vectors $\vec{E}$ and $\vec{H}$ are expressed in terms of complex quantities, the energy flow vector given by (5.1) is called the *complex Poynting vector*. In this case, however, in order to obtain real physical quantities some caution must be taken when combining complex quantities. The real energy flow in the wave is obtained by using the *real part* of both $\vec{E}$ and $\vec{H}$ in (5.1).

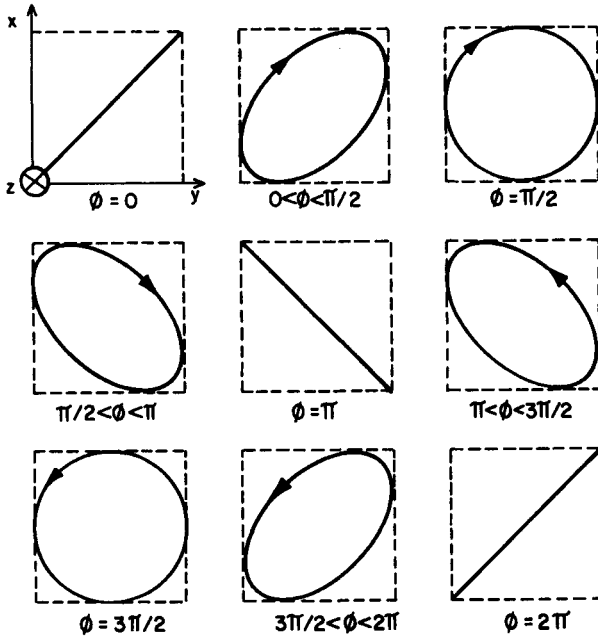


Fig. 10 - Various wave polarizations for the case when $E_x^0 = E_y^0$ and $(\alpha_2 - \alpha_1) = \phi$, obtained from (4.18) and (4.19) at the plane $z = 0$.

For the harmonic waves, the quantity of interest is the *average* energy flow (over one cycle) given by

$$\langle \vec{S} \rangle = \langle \text{Re} (\vec{E}) \times \text{Re} (\vec{H}) \rangle \quad (5.2)$$

where the time average is obtained by integrating the quantity over one period of the harmonic wave and dividing by the period, and where Re means "the real part of". It must be noted that the average energy flow $\langle \vec{S} \rangle$ is *not* given by $\langle \text{Re} (\vec{E} \times \vec{H}) \rangle$, nor by $\text{Re} \langle \vec{E} \times \vec{H} \rangle$.

In what follows we show that the time-averaged energy flow $\langle \vec{S} \rangle$, given by (5.2), can also be computed directly from the complex field vectors without performing an average over one cycle of the oscillation. For this purpose let us write

$$\vec{E} = \vec{E}_0 \exp(-i\omega t) = (\vec{E}_1 + i\vec{E}_2) \exp(-i\omega t) \quad (5.3)$$

$$\vec{H} = \vec{H}_0 \exp(-i\omega t) = (\vec{H}_1 + i\vec{H}_2) \exp(-i\omega t) \quad (5.4)$$

where $\vec{E}_1$, $\vec{E}_2$, $\vec{H}_1$ and $\vec{H}_2$ are real amplitudes, and ω is the frequency of the wave. Taking the real part of (5.3) and (5.4), yields

$$\text{Re} (\vec{E}) = \vec{E}_1 \cos(\omega t) + \vec{E}_2 \sin(\omega t) \quad (5.5)$$

$$\text{Re} (\vec{H}) = \vec{H}_1 \cos(\omega t) + \vec{H}_2 \sin(\omega t) \quad (5.6)$$

Consequently, (5.2) becomes

$$\begin{aligned} \langle \vec{S} \rangle &= \langle \text{Re} (\vec{E}) \times \text{Re} (\vec{H}) \rangle \\ &= (\vec{E}_1 \times \vec{H}_1) \langle \cos^2(\omega t) \rangle + (\vec{E}_2 \times \vec{H}_2) \langle \sin^2(\omega t) \rangle + \\ &\quad + (\vec{E}_1 \times \vec{H}_2 + \vec{E}_2 \times \vec{H}_1) \langle \sin(\omega t) \cos(\omega t) \rangle \end{aligned} \quad (5.7)$$

The following integrals are easily verified,

$$\langle \cos^2(\omega t) \rangle = \frac{1}{T} \int_0^T \cos^2(\omega t) dt = \frac{1}{2} \quad (5.8)$$

$$\langle \sin^2(\omega t) \rangle = \frac{1}{T} \int_0^T \sin^2(\omega t) dt = \frac{1}{2} \quad (5.9)$$

$$\langle \sin(\omega t) \cos(\omega t) \rangle = \frac{1}{T} \int_0^T \sin(\omega t) \cos(\omega t) dt = 0 \quad (5.10)$$

Therefore,

$$\langle \vec{S} \rangle = (\vec{E}_1 \times \vec{H}_1 + \vec{E}_2 \times \vec{H}_2)/2 \quad (5.11)$$

Let us consider now the following quantity

$$\begin{aligned} \text{Re} (\vec{E} \times \vec{H}^*) &= \text{Re} \{ [(\vec{E}_1 + i\vec{E}_2)(\cos \omega t - i \sin \omega t)] \times \\ &\quad \times [(\vec{H}_1 - i \vec{H}_2)(\cos \omega t - i \sin \omega t)] \} \\ &= \vec{E}_1 \times \vec{H}_1 + \vec{E}_2 \times \vec{H}_2 \end{aligned} \quad (5.12)$$

Comparing this result with (5.11) we obtain the following alternative form for the time-averaged energy flow (5.2)

$$\langle \vec{S} \rangle = (1/2) \text{Re} (\vec{E} \times \vec{H}^*) \quad (5.13)$$

In a medium (not free space) the time-averaged energy flow is given by

$$\langle \vec{S} \rangle = (c/4) (\vec{E} \cdot \vec{D}^* + \vec{B} \cdot \vec{H}^*) \hat{e}_3 \quad (5.14)$$

where $\hat{e}_3$ is a unit vector in the direction of $\vec{E} \times \vec{H}$.

6. WAVE PACKETS AND GROUP VELOCITY

So far we have considered simple waves having one specific value for k and ω . In practice, however, real disturbances consist of waves having some finite spread in the wave number k and in the frequency ω . A *wave packet* is a superposition of waves with different values of k and ω . This is equivalent

to the statement that any periodic motion can be decomposed by Fourier analysis into a superposition of harmonic oscillations with different frequencies ω and wave numbers k .

A wave packet, consisting of a superposition of plane harmonic waves propagating in the z direction, can be represented by

$$\psi(z, t) = \int_{-\infty}^{+\infty} A(k) \exp [i(kz - \omega t)] dk \quad (6.1)$$

The wave packet concept is particularly useful when there is only a small spread in wave numbers about a central wave number k_0 . This is equivalent to a small spread in frequencies about a central frequency ω_0 , since for any wave there is a functional relationship between k and ω , which depends on the medium, and which is called the *dispersion relation*. Thus, the amplitude function $A(k)$ is usually assumed to be peaked about some central wave number k_0 . In (6.1) the wave number k has been taken as the independent variable and ω is considered to be a function of k , determined by the dispersion relation $\omega = \omega(k)$.

The amplitude function $A(k)$ can be determined from the Fourier transform of $\psi(z, 0)$. For this purpose, we multiply (6.1) (with $t = 0$) by $\exp(-ikz)$ and integrate the resultant equation over all possible values of z . Thus,

$$\begin{aligned} \int_{-\infty}^{+\infty} \psi(z, 0) \exp(-ikz) dz &= \\ &= \int_{-\infty}^{+\infty} A(k') dk' \int_{-\infty}^{+\infty} \exp[i(k' - k)z] dz \end{aligned} \quad (6.2)$$

Using the following representation for the Dirac delta function

$$\delta(k' - k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \exp[i(k' - k)z] dz \quad (6.3)$$

and the property

$$\int_{-\infty}^{+\infty} A(k') \delta(k' - k) dk' = A(k) \quad (6.4)$$

we obtain, from (6.2),

$$A(k) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \psi(\zeta, 0) \exp(-i k \zeta) d\zeta \quad (6.5)$$

The field quantity $\psi(\zeta, t)$ can also be synthesized in terms of time periodic functions of all possible frequencies, using the relation

$$\psi(\zeta, t) = \int_{-\infty}^{+\infty} A(\omega) \exp[i(k\zeta - \omega t)] d\omega \quad (6.6)$$

where the amplitude function $A(\omega)$ gives the frequency spectrum. In this case, ω is taken as the independent variable and the dispersion relation gives $k = k(\omega)$. The amplitude function $A(\omega)$ is usually peaked about some central frequency ω_0 and it can be determined from the Fourier transform of $\psi(0, t)$,

$$A(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \psi(0, t) \exp(i\omega t) dt \quad (6.7)$$

Consider now that we have a wave packet represented by (6.1), in which the range of values of k is small and is centered about some specific wave number k_0 ,

$$k_0 - \delta k \leq k \leq k_0 + \delta k \quad (6.8)$$

If $\omega(k)$ is a slowly varying function of k , then $\omega(k)$ deviates only very slightly from its value $\omega_0 = \omega(k_0)$, and we can expand $\omega(k)$ in a Taylor series about k_0 retaining only the first two terms,

$$\omega(k) = \omega_0 + (k - k_0) (\partial\omega/\partial k)_{k_0} \quad (6.9)$$

The phase factor in (6.1) can thus be written as

$$[k\zeta - \omega(k)t] = k_0\zeta - \omega_0 t + (k - k_0) [\zeta - (\partial\omega/\partial k)_{k_0} t] \quad (6.10)$$

This allows the wave packet $\psi(\zeta, t)$ of (6.1) to be recast in the form

$$\psi(\zeta, t) = \psi_m(\zeta, t) \exp [i(k_0\zeta - \omega_0 t)] \quad (6.11)$$

where

$$\psi_m(\zeta, t) = \int_{k_0 - \delta k}^{k_0 + \delta k} A(k) \exp \{i(k - k_0) [\zeta - (\partial\omega/\partial k)_{k_0} t]\} dk \quad (6.12)$$

Therefore, the wave packet $\psi(\zeta, t)$ corresponds to a carrier wave at the frequency ω_0 , modulated by the amplitude function $\psi_m(\zeta, t)$. A typical form for a wave packet is shown in Fig. 11, at a fixed instant of time.

The phases of constant packet amplitude are given by

$$\zeta - (\partial\omega/\partial k)_{k_0} t = 0 \quad (6.13)$$

and the velocity of propagation of these planes of constant phase, $d\zeta/dt$, called the *group velocity*, is given by differentiating (6.13) with respect to time,

$$v_g = (\partial\omega/\partial k)_{k_0} \quad (6.14)$$

The group velocity can be written in terms of the phase velocity ω/k as

$$v_g = \left(\frac{\partial\omega}{\partial k} \right)_{k_0} = \left[\frac{\partial(v_{ph}k)}{\partial k} \right]_{k_0} = v_{ph_0} + k_0 \left(\frac{\partial v_{ph}}{\partial k} \right)_{k_0} \quad (6.15)$$

If the medium is such that the phase velocity is not a function of k (i.e. $\omega(k) = \text{constant} \times k$), then the group velocity is equal to the phase velocity and the medium is *nondispersive*. This is the case of free space, for which $\omega = ck$. If the phase velocity decreases with increasing k (i.e. $\partial v_{ph}/\partial k < 0$), then v_g is less than v_{ph} and the medium is said to be normally *dispersive*. If the phase velocity increases with increasing

k (i.e. $\partial v_{ph}/\partial k > 0$), then v_g is larger than v_{ph} and the medium shows what is called *anomalous dispersion*.

It should be noted that, in many cases, the phase velocity of a wave in a plasma exceeds the velocity of light c . This fact is not in conflict with the relativity theory, because an infinitely long wave train of constant amplitude does not carry information. This information, however, is contained in the modulation of the carrier wave, and this modulation travels at the group velocity, which is always less than the velocity of light.

In obtaining (6.11) and (6.12), the Taylor expansion of $\omega(k)$ was carried out only up to a term linear in $(k - k_0)$. In this case the spatial variation of the amplitude function $\psi_m(\zeta, t)$, which gives the shape of the wave packet, remains the same at any time. If the higher order terms in $(k - k_0)$ are included in (6.10), then the wave packet shape will change with time and the packet will spread out. Hence, the group velocity concept is useful only for a wave packet with a very small spread in wave numbers and frequencies.

PROBLEMS

14.1 - Derive the vector wave equations, analogous to (1.8) and (1.9) for the electric ($\vec{E}$) and magnetic ($\vec{B}$) fields, for a medium in which there is a space charge density distribution $\rho(\vec{r}, t)$ and a charge current density distribution $\vec{J}(\vec{r}, t)$.

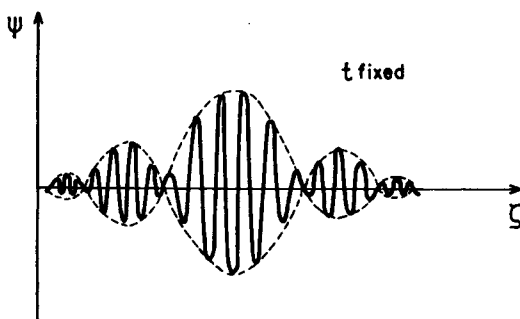


Fig. 11 - Amplitude function $\psi(\zeta, t)$ of a typical wave packet. The dotted curve is the amplitude modulation function $\psi_m(\zeta, t)$.

14.2 - Consider a plane electromagnetic harmonic wave travelling toward the positive x direction in free space, having the frequency $\nu = 5 \times 10^{15}$ Hertz.

- What is the wavelength λ in Angstroms?
- Calculate the corresponding values of ω and k .
- Considering that

$$\vec{E} = \hat{y} E_0 \exp(ikx - i\omega t)$$

calculate $\vec{\nabla} \times \vec{E}$ and $-\partial \vec{B}/\partial t$, and compare the results.

Calculate also $\vec{\nabla} \times \vec{B}$ and $\partial \vec{E}/\partial t$, and compare the results.

- Calculate the Poynting vector $\vec{S}$ and the average Poynting vector $\langle \vec{S} \rangle$.

14.3 - Consider a plane electromagnetic harmonic wave propagating along the positive x direction in free space, which can be decomposed in the sum of two waves:

$$\vec{E} = \hat{y} E_1 \exp(ikx - i\omega t) + \hat{z} E_2 \exp(ikx - i\omega t + i\alpha)$$

Show that

$$\vec{S} = \hat{x} (\epsilon_0/\mu_0)^{1/2} [E_1^2 \cos^2(kx - \omega t) + E_2^2 \cos^2(kx - \omega t + \alpha)]$$

$$\langle \vec{S} \rangle = \hat{x} (\epsilon_0/\mu_0)^{1/2} (E_1^2 + E_2^2)/2$$

14.4 - The electric field vector of an elliptically polarized plane wave, propagating in free space, can be expressed as

$$\vec{E} = [\hat{e}_1 E_1^0 \exp(i\alpha_1) + \hat{e}_2 E_2^0 \exp(i\alpha_2)] \exp(i\vec{k} \cdot \vec{r} - i\omega t)$$

- Show that the associated magnetic induction vector is

$$\vec{B} = (1/c) [\hat{e}_2 E_1^0 \exp(i\alpha_1) - \hat{e}_1 E_2^0 \exp(i\alpha_2)] \exp(i\vec{k} \cdot \vec{r} - i\omega t)$$

- Show that the time average of the energy density in this wave is

$$\langle W \rangle = (1/4) (\epsilon_0 E^2 + B^2/\mu_0)$$

Note that $E^2 = (E_1^0)^2 + (E_2^0)^2 = c^2 B^2$.

- Show that the time-averaged energy flow $\langle \vec{S} \rangle$ is equal to the phase

velocity ($\omega/k = c$) times the average energy density of the wave, that is,

$$\langle \vec{S} \rangle = c \langle W \rangle \vec{E} \times \vec{H} = (c/4) (\epsilon_0 \vec{E}^2 + \vec{B}^2/\mu_0) \vec{E} \times \vec{H}$$

14.5 - Show that in a medium (not free space), the time average of the energy density (over one cycle) for harmonic waves is given by

$$\langle W \rangle = (1/4) (\vec{E} \cdot \vec{D}^* + \vec{B} \cdot \vec{H}^*)$$

Consequently, show that the average Poynting vector (energy flow) is given by

$$\langle \vec{S} \rangle = (c/4) (\vec{E} \cdot \vec{D}^* + \vec{B} \cdot \vec{H}^*) \vec{E} \times \vec{H}$$

14.6 - Consider the superposition ($\vec{E} = \vec{E}_1 + \vec{E}_2$) of the following waves:

$$\vec{E}_1 = \hat{e}_1 E_1^0 \exp(i\alpha_1) \exp(i\vec{k} \cdot \vec{r} - i\omega t)$$

$$\vec{E}_2 = \hat{e}_2 E_2^0 \exp(i\alpha_2) \exp(i\vec{k} \cdot \vec{r} - i\omega t)$$

Analyze the resultant polarization for the following cases:

(a) $\alpha_2 = \alpha_1$ and $E_2^0 \neq E_1^0$

(b) $\alpha_2 = \alpha_1 \pm \pi/2$ and $E_2^0 = E_1^0 = E^0$

(c) $\alpha_2 = \alpha_1 \pm \pi/2$ and $E_2^0 > E_1^0$

(d) $\alpha_2 = \alpha_1 + \phi$ and $E_2^0 = E_1^0$, with $\alpha_1 = 0$ and ϕ varying from 0 to 2π

(e) Same as in (d), but with $E_2^0 > E_1^0$ and $\alpha_1 = \pi/2$.

14.7 - Generalize Eqs. (6.1) through (6.5) for the three-dimensional case in Cartesian coordinates.

14.8 - Show that the time evolution of a wave packet $\psi(\vec{r}, t)$ can be expressed in terms of the initial form of the wave packet $\psi(\vec{r}', t_0)$ as

$$\psi(\vec{r}, t) = \int_{-\infty}^{+\infty} G(\vec{r}, t; \vec{r}', t_0) \psi(\vec{r}', t_0) d^3 r'$$

where $G(\vec{r}, t; \vec{r}', t_0)$ denotes the Green function, or kernel of the integral [which depends on the dispersion relation $\omega(k)$],

$$G(\vec{r}, t; \vec{r}', t_0) = \left(\frac{1}{2\pi}\right)^3 \int_{-\infty}^{+\infty} \exp [i\vec{k} \cdot (\vec{r} - \vec{r}') - i\omega(t - t_0)] d^3 k$$

14.9 - Analyze the meaning of the Green function of Problem 14.8 for the case when $\Psi(\vec{r}, t_0)$ is given by the Dirac delta function

$$\Psi(\vec{r}, t_0) = \delta(\vec{r} - \vec{r}_0)$$

14.10 - Calculate the Green function of Problem 14.8 for the case of free space, in which $\omega = ck$, where c is the speed of light in free space. Show that, in this case, the initial wave packet $\Psi(\vec{r}, t_0)$ maintains its original shape and there is no dispersion.

14.11 - Consider a one-dimensional wave packet at the instant $t = 0$, whose amplitude function $A(k)$ is given by

$$A(k) = 1 \quad \text{for} \quad -a \leq k \leq a$$

$$A(k) = 0 \quad \text{otherwise.}$$

Show that

$$\Psi(x, 0) = 2 [\sin(ax)]/x$$

Make a plot of both $A(k)$ and $\Psi(x, 0)$, and verify that the uncertainties in x and k satisfy the relation

$$(\Delta x) (\Delta k) \geq 4\pi$$

14.12 - Consider a one-dimensional wave packet at the instant $t = 0$, whose amplitude function $A(k)$ is given by the Gaussian function

$$A(k) = \exp [-a^2 (k - k_0)^2]$$

where a and k are constants.

(a) Show that $\Psi(x, 0)$ is also a Gaussian function given by

$$\Psi(x, 0) = (\sqrt{\pi}/a) \exp (i k_0 x) \exp (-x^2/4a^2)$$

Make a plot of both $A(k)$ and $\text{Re}\{\psi(x,0)\}$, considering that $k_0 \gg 1/a$.

(b) The average extension of the wave packet $\overline{\Delta x}$ can be defined in terms of the root mean square deviation, $\overline{\Delta x} = \langle (\Delta x)^2 \rangle^{1/2}$, where $\langle (\Delta x)^2 \rangle$ is the dispersion,

$$\langle (\Delta x)^2 \rangle = \langle (x - \langle x \rangle)^2 \rangle = \frac{\int_{-\infty}^{+\infty} |\psi|^2 (x - \langle x \rangle)^2 dx}{\int_{-\infty}^{+\infty} |\psi|^2 dx}$$

Similarly, $\overline{\Delta k} = \langle (\Delta k)^2 \rangle^{1/2}$, where

$$\langle (\Delta k)^2 \rangle = \langle (k - \langle k \rangle)^2 \rangle = \frac{\int_{-\infty}^{+\infty} |A|^2 (k - \langle k \rangle)^2 dk}{\int_{-\infty}^{+\infty} |A|^2 dk}$$

Show that for this Gaussian wave packet we have

$$\overline{\Delta x} = a$$

$$\overline{\Delta k} = 1/(2a)$$

Consequently, $\overline{\Delta x} \overline{\Delta k} = 1/2$. It can be shown that the Gaussian wave packet is the minimum uncertainty wave packet and that, in general, $\overline{\Delta x} \overline{\Delta k} \geq 1/2$.

14.13 - Calculate $\psi(x,0)$ for a one-dimensional wave packet, when the amplitude function $A(k)$ is given by

(a) $A(k) = \exp(-i k x_0)$

(b) $A(k) = \delta(k - k_0)$

For both cases, verify the validity of the uncertainty principle for wave packets (which states that, in general, $\overline{\Delta x} \overline{\Delta k} \geq 1/2$).

14.14 - Consider an *evanescent* plane electromagnetic wave (for which $\vec{k} = i\alpha \vec{x}$, with α real), with the field vectors $\vec{E}$ and $\vec{H}$ proportional to $\exp(i\vec{k} \cdot \vec{r} - i\omega t)$. Show that the average value of the Poynting vector $\langle \vec{S} \rangle$ is zero for an evanescent wave.